

ABDUL MOIZ | 7 MINS READ

"Top AWS Interview Questions (With Answers) – 2025 Edition"

DEVOPS IN MOTION



◆ 1. How does AWS ensure high availability in EC2 and RDS?

Answer:

AWS provides high availability using features like:

- **EC2:** High availability is achieved using *Auto Scaling Groups*, *Elastic Load Balancers*, and *multi-AZ deployment*. You can distribute EC2 instances across multiple Availability

Zones (AZs) to ensure redundancy.

- **RDS:** For RDS, enabling *multi-AZ deployment* automatically provisions a synchronous standby replica in a different AZ. Failover is automatic in case of instance failure.

◆ 2. What's the difference between S3 and EBS?

Answer:

Feature	S3 (Simple Storage Service)	EBS (Elastic Block Store)
Type	Object storage	Block storage
Use Case	Storing files, backups, logs, etc.	Attaching to EC2 for file systems
Access	HTTP-based via API	Requires EC2 to mount
Availability	Regional	AZ-specific
Scalability	Virtually unlimited	Limited to EC2 volume type

◆ 3. How would you architect a serverless application on AWS?

Answer:

A typical serverless architecture includes:

- **API Gateway** – to expose HTTP endpoints.
- **AWS Lambda** – to run backend logic.

- **DynamoDB** – as a NoSQL database.
- **S3** – to store static files.
- **Cognito** – for authentication and user pools.
- **CloudWatch** – for monitoring and logs.

◆ 4. What is the AWS Shared Responsibility Model?

The **Shared Responsibility Model** divides security responsibilities:

- **AWS (Cloud Provider)** – Responsible for *security of the cloud*: hardware, software, networking, physical data centers.
- **Customer** – Responsible for *security in the cloud*: data, identity & access management, application-level security.

◆ 5. What are some cost optimization strategies in AWS?

Answer:

- **Use Reserved Instances or Savings Plans** for steady-state workloads.
- **Leverage Auto Scaling** to shut down resources when not needed.
- **Spot Instances** for non-critical, fault-tolerant applications.
- **Enable AWS Budgets and Cost Explorer** for visibility.
- **Use S3 lifecycle policies** to transition or delete old data.

◆ 6. How does IAM work in AWS?

Answer:

IAM (Identity and Access Management) controls **who** can access **what** in AWS:

- **Users/Roles** – Assigned to people, apps, or services.
- **Policies** – JSON documents that define permissions.
- **Groups** – For permission management at scale.
- **MFA (Multi-Factor Authentication)** – Adds security.
- **Least Privilege Principle** – Only give access needed.

◆ 7. How would you handle stateful vs. stateless applications in AWS?

Answer:

- **Stateless** apps (e.g., REST APIs) don't retain session data. You can scale them horizontally with *Load Balancers*, *Auto Scaling*, and *Lambda*.
- **Stateful** apps (e.g., session-based apps) require persistence, so you'd use *sticky sessions*, *ElastiCache*, *RDS*, or *DynamoDB* to store state.

◆ 8. ECS vs. EKS vs. Fargate – what's the difference?

- Answer:

Feature	ECS	EKS	Fargate
Type	AWS native container orchestration	Kubernetes (managed by AWS)	Serverless container runtime
Management	Fully managed by AWS	User manages Kubernetes	No server mgmt needed
Complexity	Simpler	Complex (requires Kubernetes knowledge)	Very simple



Final Thought

Knowing **AWS services** is essential. But understanding how they fit into **real-world architecture**, **cost management**, **security**, and **scalability** is what sets you apart in an interview.

- ✓ Practice whiteboarding your answers.
- ✓ Tailor responses to business use cases.
- ✓ Stay hands-on — even basic labs go a long way.

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